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Editor-in-Chief
SIAM Journal on Matrix Analysis and Applications

Dear Editor,

We submit for your consideration the manuscript:

Singular Value Interlacing for Physics-Structured Projections: Condition-Number Inheritance and the Physics-Reduction Proposition

Mathematical content. This paper proves three results in matrix analysis.

Theorem 1 establishes Cauchy-type SVD interlacing inequalities for a physics-structured projection $\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P}$, where $\mathbf{P}$ is determined by the algebraic constraints of a differential-algebraic equation (DAE) system:

$$\sigma_k(\mathbf{H}) \geq \tilde{\sigma}_k \geq \frac{\sigma_{k+n_a}(\mathbf{H})}{\sqrt{1 + \rho_Q^2}}, \quad k = 1, \dots, r.$$

The correction factor $1/\sqrt{1 + \|\mathbf{Q}\|^2}$ is new; standard Cauchy interlacing (for column-restriction by an orthonormal matrix) does not carry this term, but the physics projection $\mathbf{P}$ generally does not have orthonormal columns.

Theorem 2 derives an explicit condition-number inheritance bound:

$$\kappa(\tilde{\mathbf{H}}) \leq \kappa(\mathbf{H}) \cdot \frac{1 + \rho}{1 - \rho} \cdot \Delta_r^{-1},$$

where ρ is the algebraic coupling ratio and Δ_r is the spectral gap. When $\Delta_r \geq (1 + \rho)/(1 - \rho)$, the condition number is inherited exactly: $\kappa(\tilde{\mathbf{H}}) \leq \kappa(\mathbf{H})$.

Proposition 1 (Physics-Reduction Proposition) specialises Theorem 2 to the 6-state IBR fault-current Jacobian with $r = 2$ dynamic states, proving that $\kappa(\mathbf{H}) < 30$ implies $\kappa(\tilde{\mathbf{H}}) < 30$ for standard generator parameters.

All bounds are shown to be *sharp*: we exhibit explicit matrix families attaining each inequality.

Novelty. Prior work on SVD interlacing [?] addresses column-submatrix restrictions. The physics-structured projection $\mathbf{P} = (\mathbf{I}; \mathbf{Q})$ arising from DAE constraint elimination is a perturbation of a column restriction and requires a separate analysis. The condition-number inheritance bound and the spectral-gap criterion (Corollary 1) are new.

AMS classification. 15A18 (eigenvalues), 15A60 (norms), 65F35 (numerical linear algebra), 93B30 (identifiability).

Suggested reviewers.

- Prof. Roger Horn (University of Utah) — matrix analysis
- Prof. Ilse Ipsen (NC State University) — perturbation theory, condition numbers
- Prof. Zlatko Drmac (University of Zagreb) — SVD algorithms and perturbation bounds

- Prof. Ran Ran (Beijing Institute of Technology) — interlacing and eigenvalue inequalities

This manuscript is not under review elsewhere.

Sincerely,

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